ARPITA BHATTACHARYA

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I am a curious and efficient Masters of Mathematics student at Lund University, Sweden. I have always been interested in scientific research and a career in academia, especially in the area of Mathematics. I have taken part in multiple projects and devoted my time to study many advanced mathematical topics and concepts, mainly in Differential Equations, Analysis and other similar domains. This motivates me to continue research in this area and thus, aspire to obtain a PhD position at a prestigious institute where I can contribute meaningfully to the project's research goals.

SKILLS

Software Programming	Python, R, C++, Java	★★★☆☆
Simulation	MATLAB	★★★☆☆
ML/AI	PyTorch	★☆☆☆☆
Data Processing/Visualization	Tableau	★★☆☆☆
Documentation	Markdown, Latex, MS Office	★★★★☆

PROJECT AND RESEARCH EXPERIENCE

MASTER THESIS (Supervisor: Prof. Erik Wahlén) February 2025 – March 2026

A study of how singularities arise and develop in the surface profile, particularly when extending classical irrotational models to include constant vorticity. Involves the deep study of Plotnikov's proof of the Stokes conjecture, and Tanveer's paper 'Singularities in water waves and Rayleigh–Taylor instability', both of which concern irrotational flow. Then, make observations by including constant vorticity, one which is in its presence, one can get overhanging waves. The development of overhanging waves is related to critical layers touching the surface from outside the fluid domain, assuming the velocity field is continued there in an appropriate way.

PROJECTS (MASTER AND BACHELOR COURSES) September 2023 – January 2025

Numerical Methods for Partial Differential Equations (Under guidance of: *Prof. Robert Klöfkorn*, Lund University, Sweden)
Project tasks based on Elliptic partial differential equations and investigating its various properties using DUNE & DUNE-FEM in Python
Project 1, Project 2 and Project 3.

Numerical Methods for Differential Equations (Under guidance of: *Prof. Tony Stillfjord*, Lund University, Sweden)
Project tasks completed using MATLAB and presented to the TAs.
Tasks consisted of concepts: Euler method, Adaptive Runge-Kutta methods, Lotka-Volterra equation, Van der Pol problem, 2-point BVP solver, Beam equation, Strum-Liouville eigenvalue problem, Schrödinger equation, Diffusion equation, Advection equation, Convection diffusion equation, Viscous Burgers' equation.
Presentation 1, Presentation 2 and Presentation 3

Mathematical Modelling (Under guidance of: *Prof. Joachim Hein*, Lund University, Sweden)
Project tasks completed using Python and Microsoft Excel and presented to the Professor.
Project 1: Optimization project: Textile distribution
Project 2: Analysis of Predator-prey modelling

Numerical Linear Algebra
Assignment tasks completed using Python
Tasks consisted of concepts: Gram-Schmidt orthogonalization, QR-factorization, Householder transformations, Givens rotations, Image Compression using SVD, Hilbert Matrices, LUdecomposition, hyperlink matrices, GMRES.
Project 1, Project 2 and Project 3.

ElGamal Cryptography

Used Number Theory concepts and Python to code the algorithm.
Project Report

Elliptic Curve Cryptography (Under guidance of: *Dr. Weiyi Zhang*, University of Warwick, United Kingdom)
Writing research essay on the topic using concepts in Abstract Algebra and Number Theory.
Project research paper.

SUBJECT MATTER EXPERT (FREELANCE) — February 2023 – April 2024

Chegg (Advanced Mathematics, *Managed Network Expert*, Bengaluru, India)
I contributed and shared my mathematical knowledge by answering and discussing questions asked by students ranging from high school to those already graduating from university.

EDUCATION

Master in Mathematics (Numerical Analysis) Lund University, Lund, Sweden	September 2023 – March 2026	Grade: G
Bachelor in Mathematics University of Warwick, Coventry, United Kingdom	September 2019 – August 2022	Grade: Hons.
National Public School (High School) Bengaluru, Karnataka, India	June 2005 – March 2019	Final Score: 91%
Centre of Fundamental Research and Creative Education (CFRCE), (Research) Bengaluru, Karnataka, India	April 2017 – August 2019	GPA: 3.7

EXTRA-CURRICULAR

International Student Ambassador
September 2023 – March 2026
Lund University, Lund, Sweden

- Responsibility of a number of tasks designed to help prospective students in their decision-making process.
- Being a representative of Lund University.
- Answering questions and providing information via the Unibuddy platform and Live events
- Creating content for blogs and social channels of the university

Company Host at ARKAD (Career Fair)
October 2023
Lund University, Lund, Sweden

- Point of contact between ARKAD and two companies in career fair. Help order resources & materials, set up booths, personally assist the company representatives during the fair

Warwick Student Volunteers at University of Warwick, Coventry, UK
January 2020 – March 2020
University of Warwick, Coventry, United Kingdom

- Helped primary school children learn mathematics through fun games, building their confidence in math concepts and making it fun as part of project ‘Fun with numbers’.

National Centre for Human Settlements and Environment (NCHSE)
April 2018
Madhya Pradesh, India

- “Integrated Watershed Management Programme”- Survey of health, hygiene and sanitation awareness in project villages, in the districts of Madhya Pradesh, India.

OTHER INTERESTS/HOBBIES

Music

INSTRUMENT

- Sitar – Hindustani Classical

DANCE

- Hindustani Classical – Kathak, Bharat Natyam
- Contemporary – Street

Reading

FICTION, SCIENCE, TECHNOLOGY, PHILOSOPHY